

Final Scientific Validation Report

Experiment ID: P6-FINAL-VALIDATION **Date:** 2026-07-18

Executive Summary

This report presents the final scientific validation of the **LigProtGNN-X** project. We verified the biological and computational correctness of the cross-attention and geometry-attention GNN modules. All scientific hypotheses (H4.1 through H5.5) have been validated and confirmed.

- **Hypothesis Status:** Validated
- **Regression Analysis:** Checked. No performance regression was introduced by any late-stage integration.